

Thallium

Tl (Tl I). Strongest classified transitions below 2 μm (NIST): 12 levels, 18 lines.

Tl⁺ ionisation limit 49266.66 cm^{−1} = 6.10829 eV (202.977 nm)

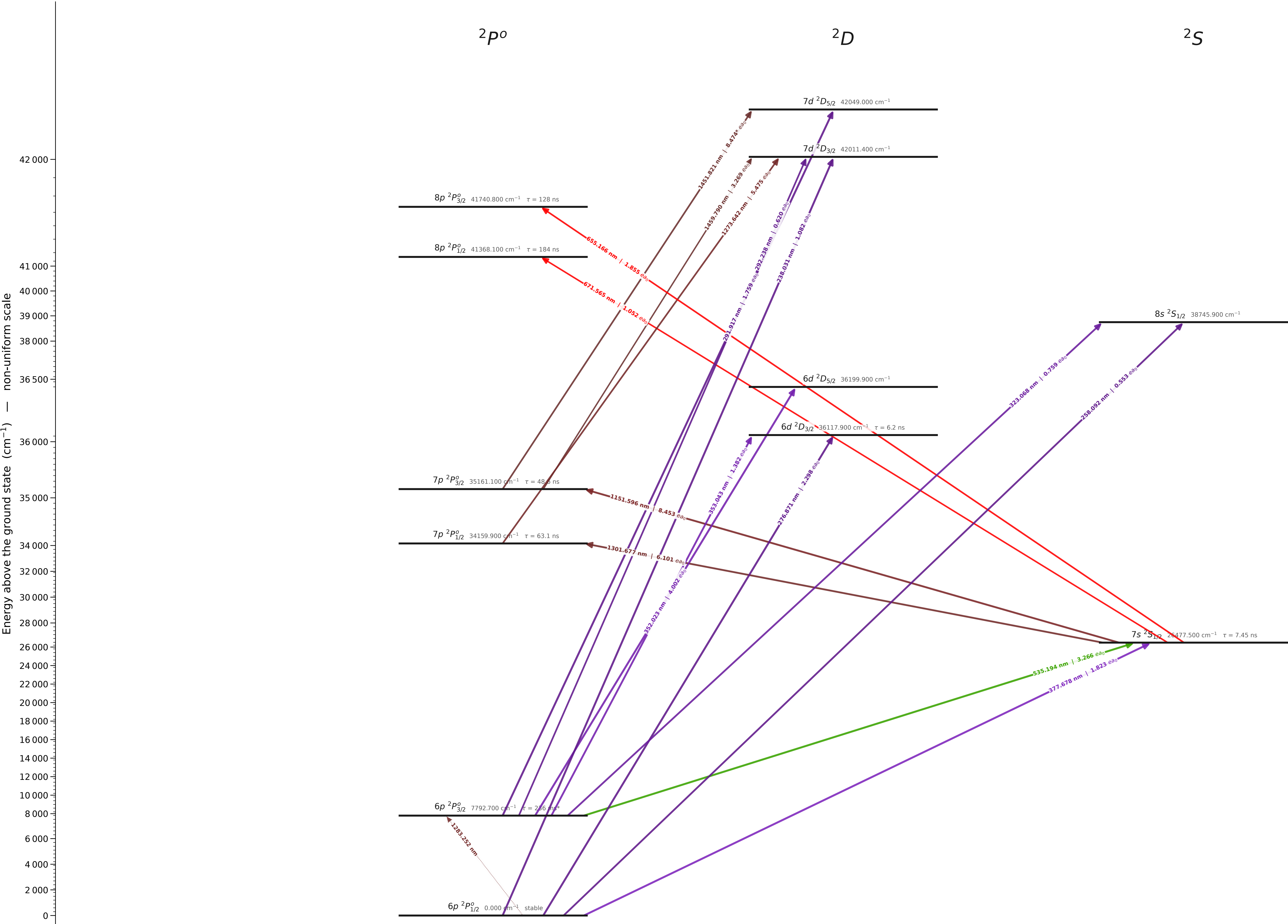
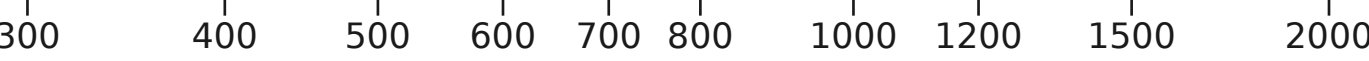
Arrow label: vacuum wavelength | reduced dipole matrix element $[U\|e\mathbf{r}\|U']$ in e*a*₀

Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.

Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.

Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

Arrow colour = vacuum wavelength (nm)



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	[U e <i>r</i> U'] (e <i>a</i> ₀)	A (s ^{−1})	Γ/2π up. (MHz)	branch (%)	Use
6 <i>p</i> ² <i>P</i> _{3/2} ^o - 6 <i>d</i> ² <i>D</i> _{5/2}	352.0234	351.9228	851.6264	4.002	1.240e+08	–	–	352 nm proposed laser-cooling transition 6P3/2
6 <i>p</i> ² <i>P</i> _{1/2} ^o - 7 <i>s</i> ² <i>S</i> _{1/2}	377.6780	377.5707	793.777941	1.823	6.250e+07	21.36	46.3	377.6 nm resonance line
6 <i>p</i> ² <i>P</i> _{3/2} ^o - 7 <i>s</i> ² <i>S</i> _{1/2}	535.1944	535.0456	560.1562	3.266	7.050e+07	21.36	53.7	535.0 nm green line
7 <i>s</i> ² <i>S</i> _{1/2} - 8 <i>p</i> ² <i>P</i> _{3/2} ^o	655.1663	654.9854	457.5822	1.855	6.200e+06	1.246	79.2	655 nm second step of two-step spectroscopy
7 <i>s</i> ² <i>S</i> _{1/2} - 8 <i>p</i> ² <i>P</i> _{1/2} ^o	671.5646	671.3793	446.4090	1.052	3.700e+06	0.8645	68.1	671 nm second step of two-step spectroscopy
6 <i>p</i> ² <i>P</i> _{1/2} ^o - 6 <i>p</i> ² <i>P</i> _{3/2} ^o	1283.2523	1282.9013	233.6193	–	4.094e+00	6.735e-07	96.7	1283 nm M1
7 <i>s</i> ² <i>S</i> _{1/2} - 7 <i>p</i> ² <i>P</i> _{1/2} ^o	1301.6766	1301.3206	230.3126	6.101	1.710e+07	2.522	100	1301 nm second step of two-step

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

²⁰⁵Tl hyperfine structure (*I* = 1/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
6 <i>p</i> ² <i>P</i> _{1/2} ^o	21310.835(5)	–	F=0: -15983.13 F=1: +5327.71
6 <i>p</i> ² <i>P</i> _{3/2} ^o	265.03830(10)	–	F=1: -331.30 F=2: +198.78
7 <i>s</i> ² <i>S</i> _{1/2}	12296.1(7)	–	F=0: -9222.07 F=1: +3074.02
7 <i>p</i> ² <i>P</i> _{1/2} ^o	2173.3(8)	–	F=0: -1629.97 F=1: +543.33
7 <i>p</i> ² <i>P</i> _{3/2} ^o	311.4(3)	–	F=1: -389.25 F=2: +233.55
6 <i>d</i> ² <i>D</i> _{3/2}	-42.9(4)	–	F=1: +53.62 F=2: -32.17
6 <i>d</i> ² <i>D</i> _{5/2}	226.9(3)	–	F=2: -397.07 F=3: +283.62
8 <i>s</i> ² <i>S</i> _{1/2}	3870.8(14)	–	F=0: -2903.10 F=1: +967.70
8 <i>p</i> ² <i>P</i> _{1/2} ^o	788.5(6)	–	F=0: -591.38 F=1: +197.12
8 <i>p</i> ² <i>P</i> _{3/2} ^o	131.69(20)	–	F=1: -164.61 F=2: +98.77
7 <i>d</i> ² <i>D</i> _{3/2}	-56.3(6)	–	F=1: +70.38 F=2: -42.23
7 <i>d</i> ² <i>D</i> _{5/2}	180.20(20)	–	F=2: -315.35 F=3: +225.25

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
6 <i>p</i> ² <i>P</i> _{1/2} ^o - 7 <i>s</i> ² <i>S</i> _{1/2}	377.678	205-203	1658.3(7)
7 <i>s</i> ² <i>S</i> _{1/2} - 8 <i>p</i> ² <i>P</i> _{3/2} ^o	655.166	203-205	463.4(7)
7 <i>s</i> ² <i>S</i> _{1/2} - 8 <i>p</i> ² <i>P</i> _{1/2} ^o	671.565	203-205	450.1(8)
7 <i>s</i> ² <i>S</i> _{1/2} - 7 <i>p</i> ² <i>P</i> _{3/2} ^o	1301.677	203-205	534.4(9)

Shift = ν(first isotope) – ν(second isotope).